Substitution to the Subway?

A basic statistical analysis of the MTA in NYC

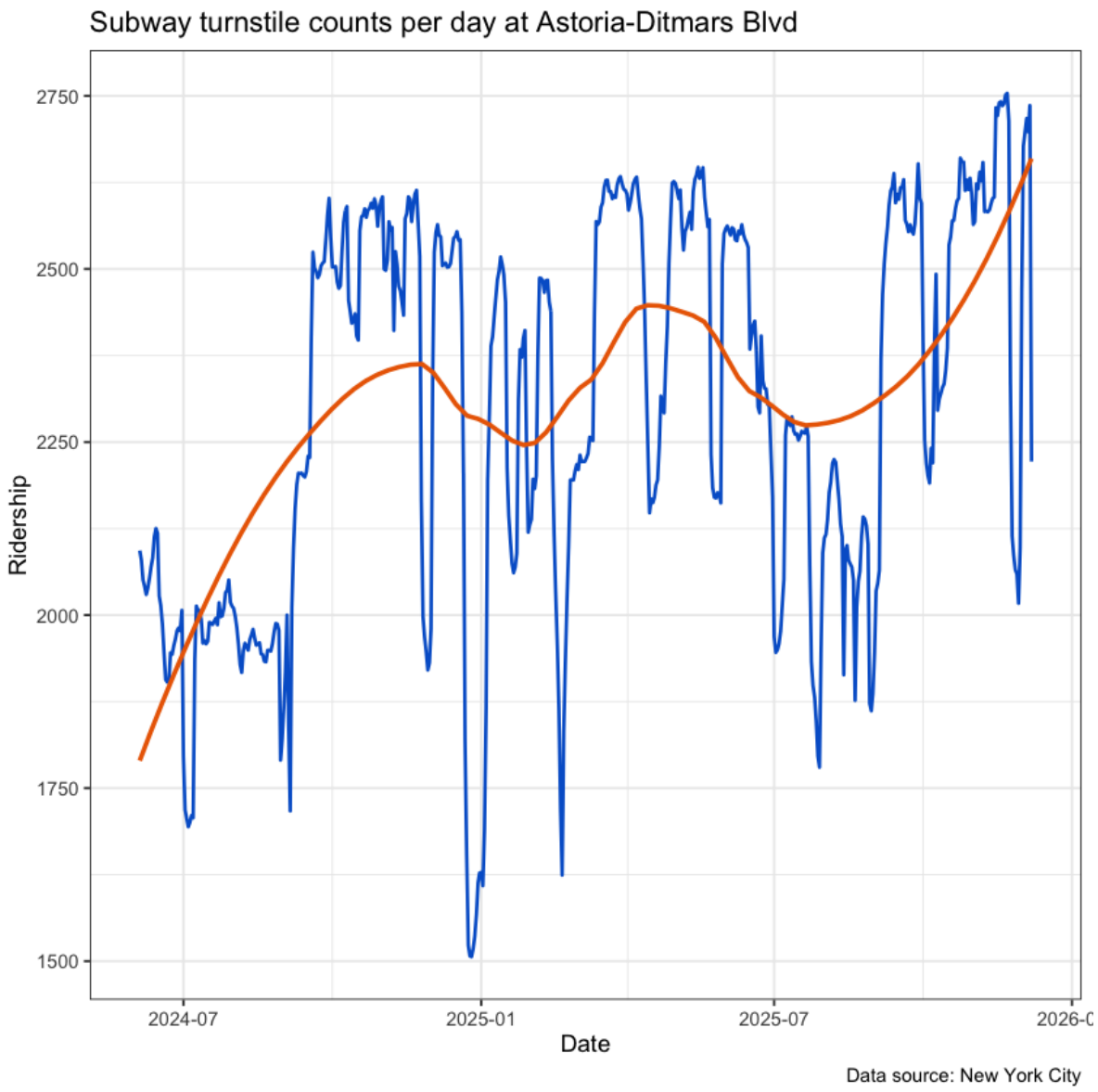
STAT5010 | Spring 2025 | Lily Holmes

Motivation

Transportation is vital, especially in a densely populated area

What is being analyzed is a think that impact real people

Future policy + precedents



Background

- $9 tax imposed to enter the the Central Business District in lower Manhattan during peak hours
- Goals and outcomes already studied
  - Traffic reduction
  - Improve air quality
  - Raise revenue

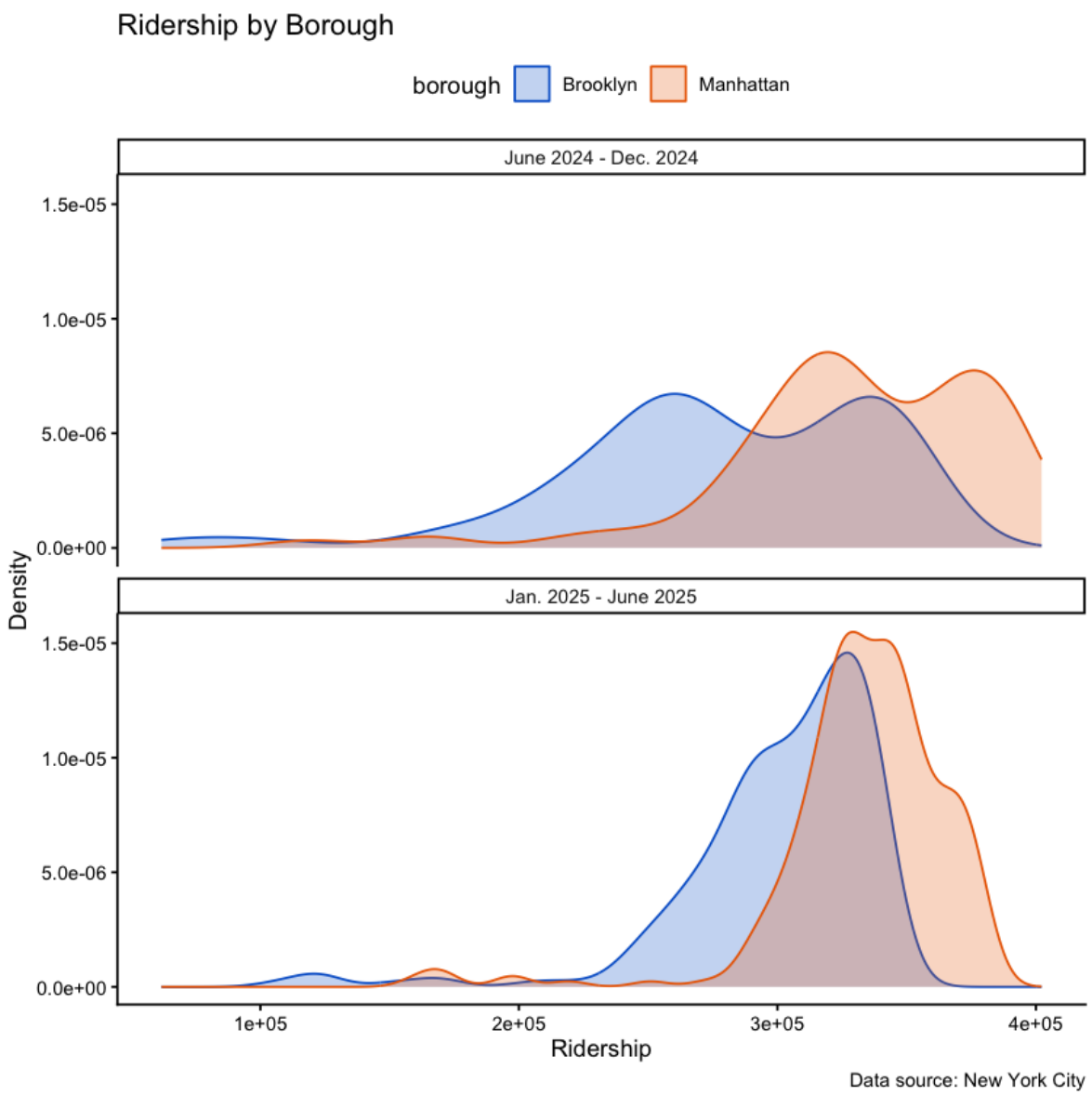
photo credits: https://www.nycsubway.org/wiki/New_York_City_Subway_Route_Map_by_Michael_Calcagno

Did the infliction of the NYC congestion tax for passageways leading into the CBD have a direct result on the transportation decisions made?

Research Goals

1. Understand what real world, measurable, factors impact daily ridership, and to what extent at what level

2. Isolate and interpret the potential treatment effect using causal methods



Data

- MTA Hourly Subway Counts
  - Two sets combined
    - 2020-2025
    - 2025 onward
  - For regression
    - 1. Filtered to isolate single station
    - Or just aggregated for daily counts
  - For DiD
    - filtered for rush hour, then aggregated daily
- NOAA weather – JFK Airport
  - Rain, snowfall, snow depth, daily temperature for regression variables

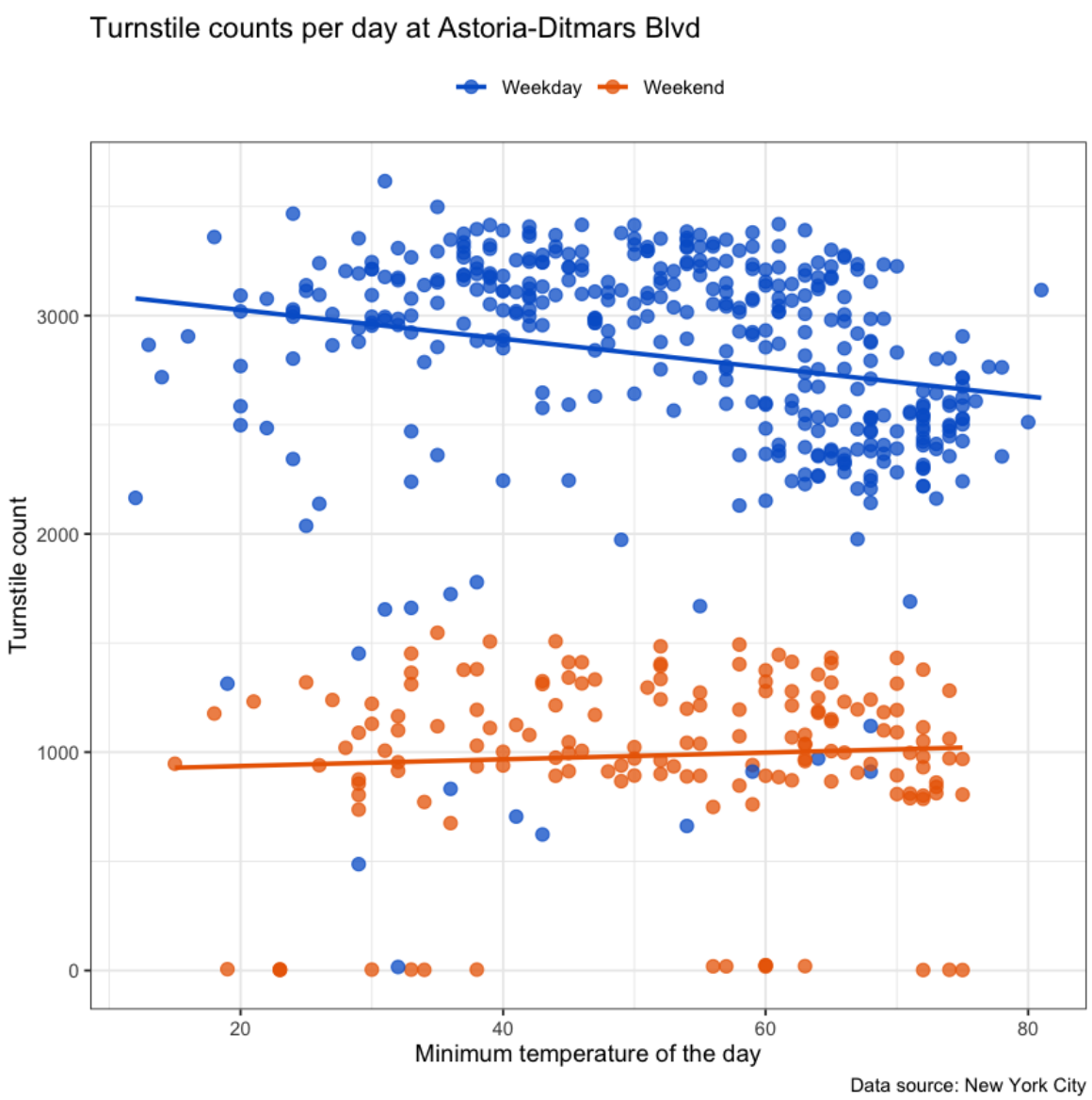
Photo credit: https://www.istockphoto.com/photo/subway-turnstiles

Multiple Linear Regression

	ridership			
	(1)	(2)	(3)	(4)
temp	−694.049	7,662.618**	7,597.892***	7,498.474***
	(496.756)	(3,199.986)	(843.851)	(848.580)
Weekend	−241,728.300***	−241,743.200***	−242,292.300***	−267,716.700***
	(17,133.830)	(17,110.850)	(4,508.232)	(15,759.410)
rain	−8,807.306	−9,768.538		−9,387.935**
	(16,465.920)	(16,447.860)		(4,339.660)
$temp^2$		−84.620***	−84.200***	−84.409***
		(32.012)	(8.444)	(8.436)
Brooklyn			463,025.900***	463,025.900***
			(5,766.526)	(5,758.717)
Manhattan			946,223.700***	946,223.700***
			(5,766.526)	(5,758.717)
Queens			218,908.000***	218,908.000***
			(5,766.526)	(5,758.717)
temp × weekend				496.301*
				(288.543)
Constant	737,413.100***	552,830.700***	144,809.000***	153,636.900***
	(27,701.960)	(75,107.460)	(20,125.100)	(20,613.560)
Observations	2,232	2,232	2,232	2,232
R^2	0.083	0.086	0.936	0.937
Adjusted R^2	0.082	0.085	0.936	0.936
Resid. Std. Error	365,535.500	365,045.300	96,319.970	96,189.530
Resid. Std. Error df	2228	2227	2225	2223
F Statistic	67.549***	52.545***	5,463.581***	4,109.812***
F Statistic df	3; 2228	4; 2227	6; 2225	8; 2223

***p<0.01, **p<0.05, *p<0.1

Note: We obtain the same adj.R2 for model 4 as we do without including the rain boolean or the interaction term



Difference in Differences with fixed effects

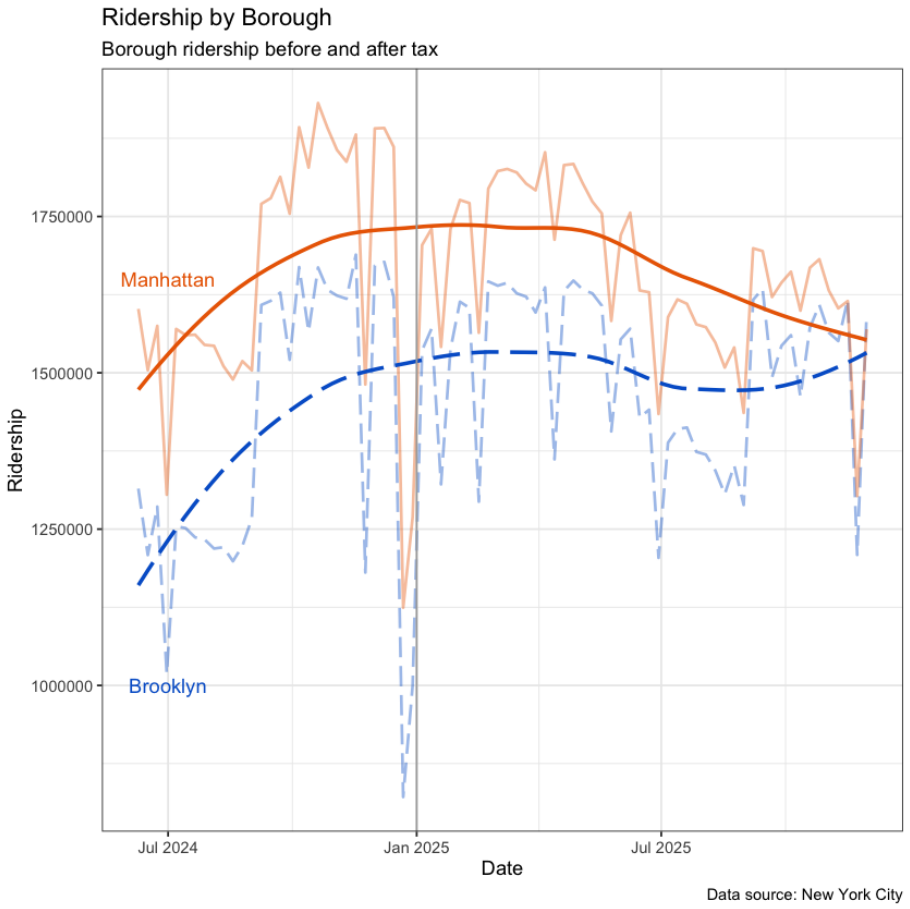


Table 3: Regressio

	Real
Manhattan x Treated	-20206.267***
	(1447.835)
Manhattan x Fake Period	
Queens x Treated	
Num.Obs.	794
R2	0.981
R2 Adj.	0.961
RMSE	7002.25
FE: borough	X
FE: date	X

Difference in Differences – Placebo Tests

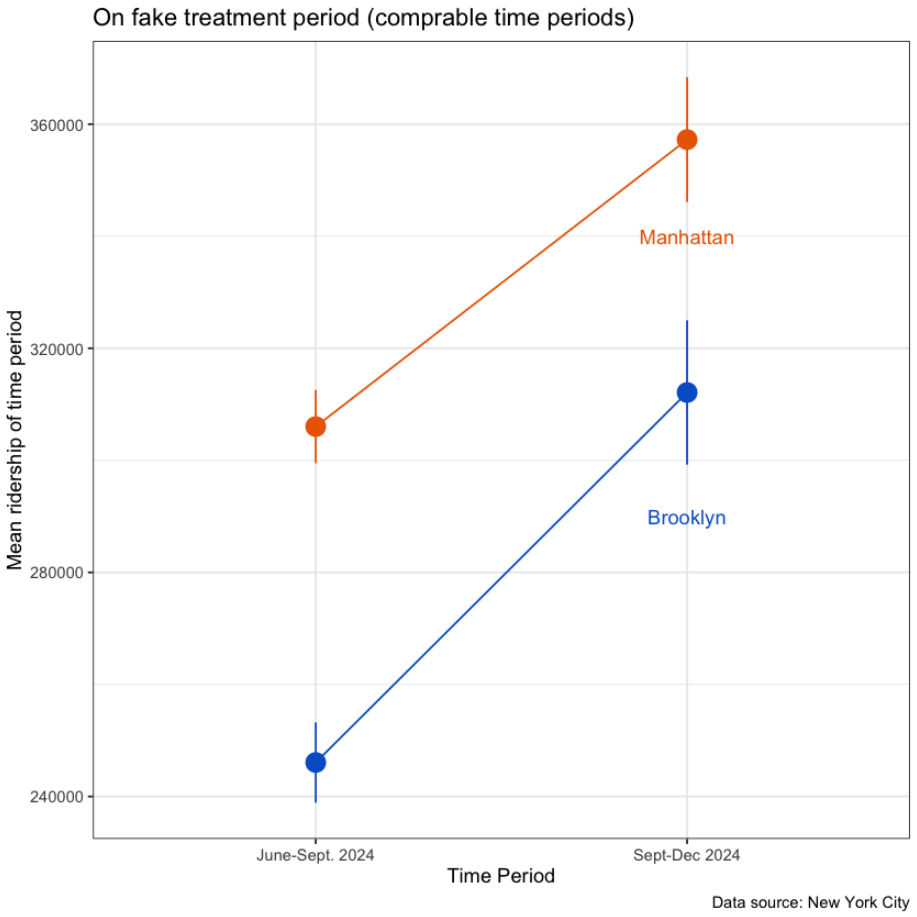
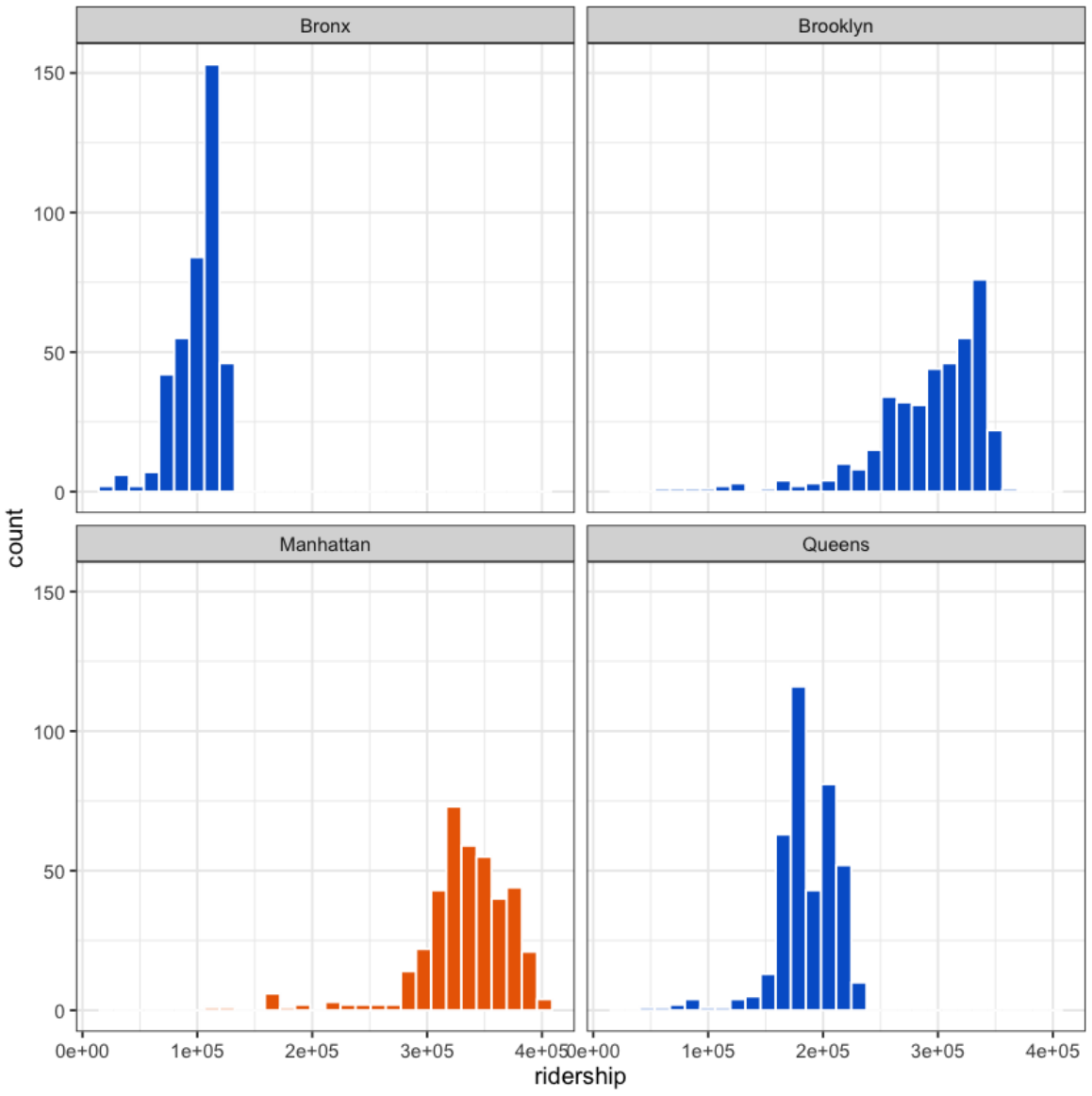


Table 3: Regression Results

	Real	Placebo Time	Placebo Borough
Manhattan x Treated	-20206.267***		
	(1447.835)		
Manhattan x Fake Period		-24166.208***	
		(1983.934)	
Queens x Treated			-25607.514***
			(2644.938)
Num.Obs.	794	794	794
R2	0.981	0.979	0.963
R2 Adj.	0.961	0.958	0.926
RMSE	7002.25	7295.09	12791.88
FE: borough	X	X	X
FE: date	X	X	X

Conclusion

- Transportation is complex
  - Impacts people's daily life and is impacted by people's daily life
- Future questions
  - What other outcomes (safety or feedback) can we predict using ridership?
  - Would the use of other methods, explanatory or predictory, show similar results?